Genetic counselling & consent
A published Indian case: diagnosis, a new pregnancy and an uncertain extra finding

Summary for doctors and healthcare teams, written in our own words from a published, peer-reviewed case report. Please read the original article (link below).

Source: Gupta A, Sharma Y, Deo K, Vellarikkal S, Jayarajan R, Dixit V, Verma A, Scaria V, Sivasubbu S. *Case Report: Whole exome sequencing helps in accurate molecular diagnosis in siblings with a rare co-occurrence of paternally inherited 22q12 duplication and autosomal recessive non-syndromic ichthyosis.* F1000Research 2015;4:446. Dr. D.Y. Patil Medical College, Pune, and CSIR-Institute of Genomics and Integrative Biology, Delhi. Open access (CC BY 4.0). f1000research.com/articles/4-446/v1

The patients

- Two siblings, aged 8 and 6, non-consanguineous parents; no family history of skin disease
- Born in a collodion membrane, shed within a week
- Hyperpigmented fish-like scales; ectropion, eclabium; scalp-margin alopecia
- Heat intolerance, hypohidrosis, photosensitivity; bilateral keratitis

Genetic findings

- Earlier karyotype: **22q12 duplication** in the father and both children; no cat eye syndrome features in any of them
- Exome (both siblings): **homozygous TGM1 p.Gly218Ser**; both parents heterozygous carriers (Sanger)
- Diagnosis: **lamellar ichthyosis**, autosomal recessive
- Low-dose oral isotretinoin: ectropion, eclabium and alopecia resolved over two years

Why counselling mattered in this family

The mother was pregnant again. With both parents carriers, **each pregnancy carries a 25% risk**; the authors planned prenatal testing of the fetus using the identified variant. The family also needed a clear explanation of the **paternally inherited 22q12 duplication**, whose link to the skin disease was uncertain (reviewers questioned its relevance). This is the work of pre- and post-test counselling: recurrence risk, prenatal options with proper consent, and honest communication about uncertain findings.

Talking points for the doctor visit

- Offer pre-test counselling and take written consent before every genetic test.
- Record choices on secondary findings, sample/data storage and report sharing.
- Prenatal testing in India: written consent under the PCPNDT Act; testing for disease only; the sex of the fetus is never disclosed.
- After the report: recurrence risk, cascade testing of relatives, referral and emotional support; no major decisions on a VUS.

For healthcare professionals' general awareness. Not a substitute for clinical judgement or legal advice; facts and guidelines may change. Case details are as published by the original authors; consult the full article.

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Genetic counselling & consent checklist

Stage	Cover
Pre-test counselling	Why this test; what it can and cannot detect; possible results (positive, negative, VUS); turnaround and cost; non-directive approach (the family decides).
Consent decisions	Report secondary findings or not; store sample/data for re-analysis; who may receive the report; written consent signed by patient or guardian.
Family implications	Risk to siblings, children and parents; trio testing can reveal unexpected family-relationship information (discuss before consent).
Children	Predictive testing for adult-onset conditions with no childhood intervention is generally deferred until adulthood (ASHG 2015).
Prenatal (India)	PCPNDT Act: written consent; disease testing only; fetal sex never disclosed.
Post-test counselling	Explain the report simply; management and referrals; recurrence risk; cascade testing; emotional support; no major decisions on a VUS.
Privacy	Genetic information is sensitive; keep reports confidential and share only with consent (ICMR National Ethical Guidelines, section on human genetics testing).

References

- Gupta A et al. F1000Research 2015;4:446. f1000research.com/articles/4-446/v1
- ASHG position statement: Points to consider: ethical, legal and psychosocial implications of genetic testing in children and adolescents. Am J Hum Genet 2015. summary (Healio)
- ICMR National Ethical Guidelines for Biomedical and Health Research Involving Human Participants, 2017 (section 10: human genetics testing and research). PDF
- Pre-Conception and Pre-Natal Diagnostic Techniques (PCPNDT) Act, 1994 (India).

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